



Ripha College Of Science & Technology

Fall – 2025

Semester 7

FYP Proposal

Title: *Online Bus Tracking & Management System*

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Contents

1. Problem Statement:-----	3
2. Objectives of Project:-----	3
I. To provide real-time tracking of university buses-----	3
II. To reduce waiting time and prevent students from missing buses-----	3
III. To improve the safety and monitoring of students during travel -----	3
3. Methodology:-----	4
i. Requirement Analysis-----	4
ii. System Design: -----	4
iii. Software Development-----	4
iv. Deployment & Maintenance-----	5
4. Activity Diagram -----	5
5. Survey Form Review: -----	6
6. Budget:-----	8
7. Project Proposed Time Schedule: -----	8
8. Conclusion:-----	8
9. Supervisor's Comments: -----	9

1. Problem Statement:

Students often face difficulties because they do not know the exact location or arrival time of their university bus, which leads to delays, long waiting times, and missed buses. At the same time, managing student fees manually creates errors, confusion, and extra workload for the administration. No combined system can track buses in real time and manage student fee records accurately.

2. Objectives of Project:

I. To provide real-time tracking of university buses

The main objective is to allow students, parents, and university staff to view the bus location at any moment through a Mobile application. Real-time GPS data helps students to plan their university schedule and reduces uncertainty. This feature also ensures transparency and improves the reliability of the transportation system.

II. To reduce waiting time and prevent students from missing buses

Many students wait unnecessarily because they do not know when the bus will reach their stop.. This saves time, reduces frustration, and makes daily travel more convenient.

III. To improve the safety and monitoring of students during travel

By tracking the bus live, parents and university security can monitor the bus route and status, and in case of delays, emergencies, or unusual route changes, the system can alert the administration. This increases the overall safety of the students and builds trust in the transport system.

IV. To create a secure online fee management system

In the multiple features of this application, one is fee management, which manages all transport-related fee requirements through this application and maintains states of punctuality.

3. Methodology:

Implementation of this project, we consider it in steps.

i. Requirement Analysis

Collect requirements from the University for students, parents, and bus staff. Identify features like:

- Bus tracking,
- Real-time notifications,
- Fee management.

Bus tracking:

Bus tracking is a system that shows the live position of the bus on a map using GPS.

Real-time notification:

Real-time notification means the system automatically sends updates when the bus reaches.

Fee management:

Include modules for student registration, fee submission, payment tracking, and receipts. Ensure secure access to staff, parents, and admin.

ii. System Design:

The online bus tracking system uses a client-server design, where the mobile app works as a client, and the main database acts as the server. The bus sends its location to the server through GPS, and the server processes this information. The mobile app then takes the updated data from the server and shows the live bus location to the user.

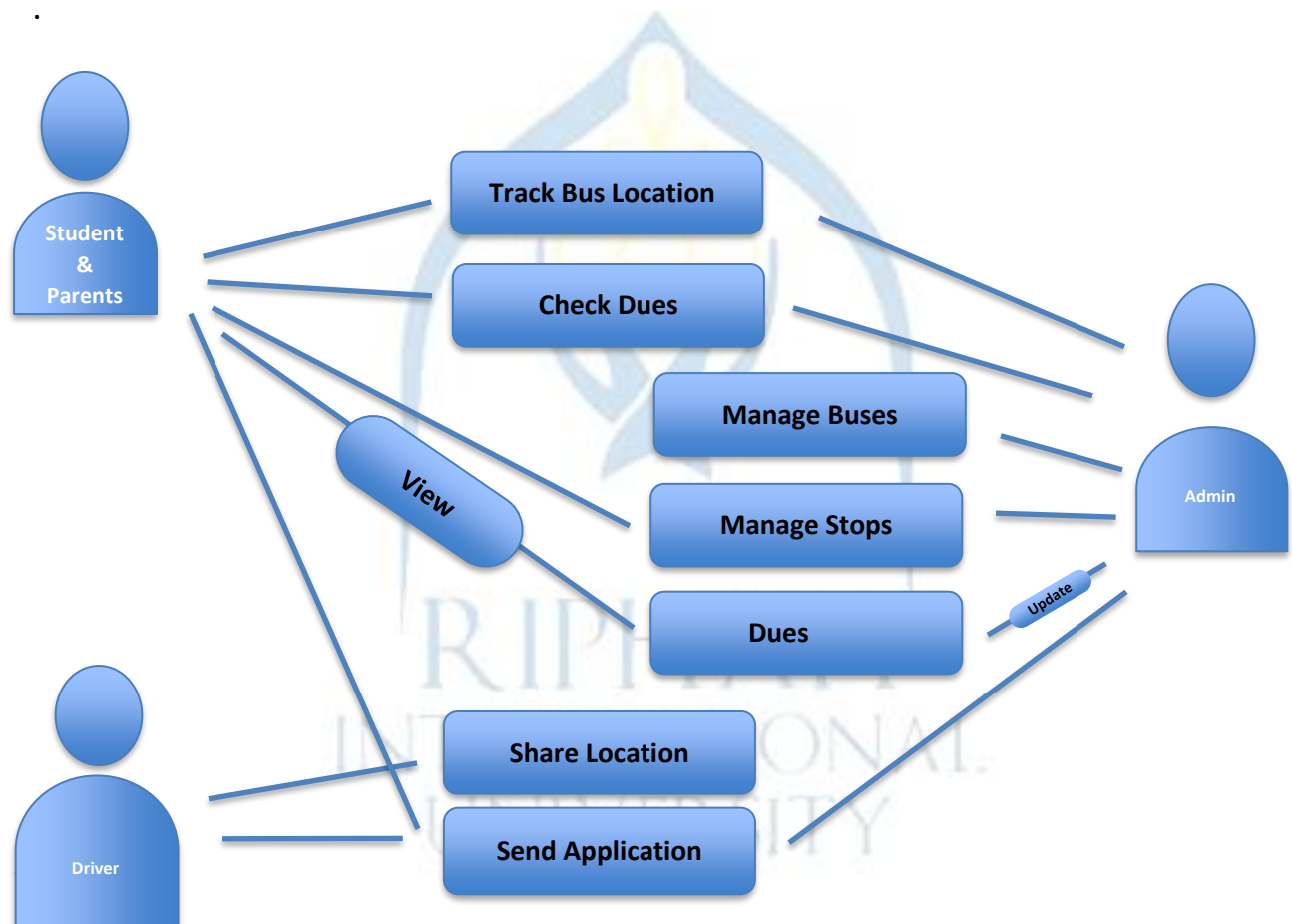
iii. Software Development

Develop a mobile application for students/Parents and University staff with an integrated database to store student information, bus routes, and fee records. Implement a notification system to alert parents about bus arrival and delays.

iv. Deployment & Maintenance

Deploy the system in schools. Regularly update software and maintain hardware to ensure smooth operation.

4. Activity Diagram

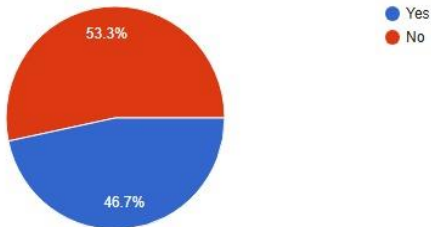


5. Survey Form Review:

Are you using University transport or not.

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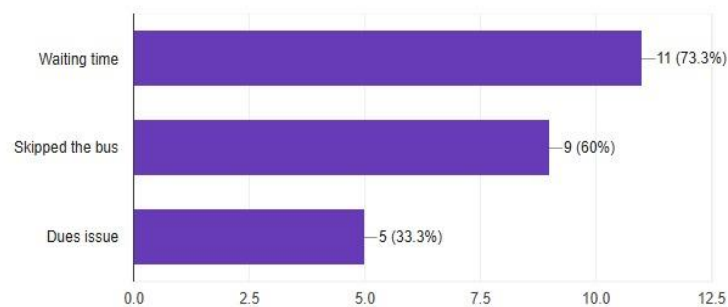
15 responses



Which type of problems your are facingin convantional bus system of university.

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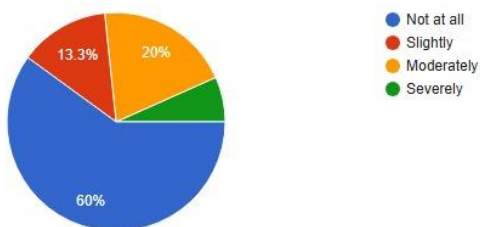
15 responses



How do delays in bus arrivals affect your routine?

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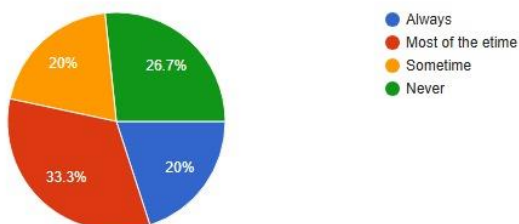
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Do you feel safe in convantional buss system while traveling for university?

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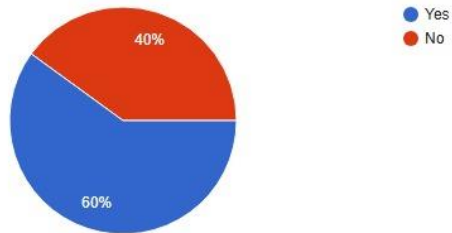
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Is the convational bus system easy to navigate (routes, schedules)?

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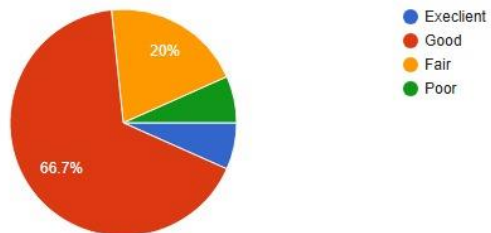
15 responses



How would you rate the behavior and helpfulness of bus staff?

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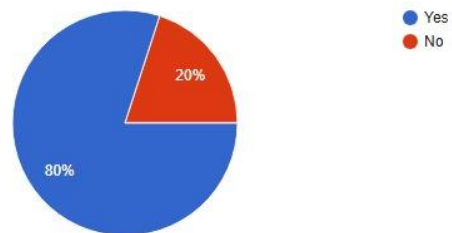
15 responses



Are you intrested to get a mobile application which handle all issues that mention before.

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15 responses



6. Budget:

Marketing	20k
Update budget	50k (4- years)
Development	50k
Operational cost	10k (per month)

7. Project Proposed Time Schedule:

	November	December	January	February	March
Requirements					
Discussion					
Working					
System design					
Software development					
Deployment and maintenance					

8. Conclusion:

This system is very helpful for students, parents, and the university. With online bus tracking, students can see where the bus is and when it will arrive, so they can reach the stop on time and avoid waiting. The digital fee system makes it easy to keep fee records, reduces mistakes, and keeps things organized. In simple words, this system makes travel easier and safer, improves management, and gives the university an easy and reliable way to handle bus and fee records.

9. Supervisor's Comments:

(To be filled by Supervisor)

Signature: _____

Name: Mr. Ameer Hamza

Designation:

Final Year Project Coordinator Remarks:

